

COI-ATI Short answer Question

1] what is Data science? Explain its Primary objective.

A] Data science is the field of extracting useful knowledge and insights from data using scientific methods, algorithms and statistics.

2] why is Data science important in today's digital world? mention any four reasons.

A] → helps in better decision-making
→ Predicts future trends.
→ Improves business efficiency
→ supports automation and innovation.

3] list any four benefits of Data science in business and society.

A] → Better customer understanding
→ Increased business profits.
→ Improved healthcare services
→ Fraud detection and prevention.

4] Mention Four real-world application of Data science.

A] → healthcare
→ Finance
→ E-commerce
→ Transportation

5] what are the different facets of data? Briefly explain each.

A] → structured data: Organized in tables
eg:- database, records.

→ Semi-structured Data: Partially organized

Eg:- XML, JSON.

→ Unstructured Data: No Fixed format

Eg:- image, videos.

6.] differentiate between structured, semi-structured and unstructured data with suitable examples.

A.] Data Type	Description	Example
structured	organized in rows and columns	student data - base
semi-structured	has tags but no fixed table	JSON, XML
unstructured	No predefined Format	Image, videos, emails.

7.] what is Big Data? state its key characteristics

A.] Large and complex data that cannot be processed using traditional methods.

→ volume → variety → value
→ velocity → veracity

8.] Explain the Big Data ecosystem. Name any four used

A.] Big Data ecosystem is a collection of tools used to store, process, and analyze large datasets.

→ hadoop → Hive
→ spark → HBase

Q] what are the major phases of the Data science process?

A] → Data collection → model Evaluation,
→ Data cleaning → Deployment
→ Data Analysis → visualization.
→ model Building

10] what is data retrieval? mention two common sources.

A] Data retrieval is the process of obtaining data from storage.
sources.

→ Databases
→ APIs/websites.

11] why is data collection considered an important step in the Data science process?

A] accurate and relevant data leads to accurate analysis and better predictions.

12] what is data cleansing? list any four common data cleaning techniques.

A] Data cleansing is the process of correcting or removing incorrect data.

→ Remove duplicates → handle missing values
→ correct errors → remove outliers.

13] what is data integration? why is it necessary in Data science?

A] Data integration combines data from multiple source into one dataset.

need: Improves data quality and Provides a Complete view.

14] Explain data transformation with suitable examples

A] Data transformation converts data into a suitable format for analysis.

Ex:- Converting date "16/07/2026" to "2026-07-16"

15] what is data analysis? mention any four techniques used for data analysis.

A] Data analysis is the process of examining data to find useful information.

→ Regression

→ clustering

→ classification

→ Data visualization

94/30

90%
11/7/26